

Question:

A 5-year-old boy is brought to the physician by his mother because he does not “listen to her” anymore. The mother also reports that her son cannot concentrate on any tasks lasting longer than just a few minutes. Teachers at his preschool report that the patient is more active compared to other preschoolers, frequently interrupts or bothers other children, and is very forgetful. Last year the patient was expelled from another preschool for hitting his teacher and his classmates when he did not get what he wanted and for being disruptive during classes. He was born at term via vaginal delivery and has been healthy except for 3 episodes of acute otitis media at the age of 2 years. He has met all developmental milestones. His mother has major depressive disorder and his father has Graves' disease. He appears healthy and well nourished. Examination shows that the patient does not seem to listen when spoken to directly. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in treatment?

Answer Choices:

- A. Behavior therapy
- B. Methimazole
- C. Fluoxetine
- D. Hearing aids

Reasoning + Answer:

<think>

Finding Reasoning Paths:

1. **Hyperactivity and poor concentration** → Suggests attention-deficit/hyperactivity disorder (ADHD) → Consider stimulant therapy (e.g., methylphenidate) or non-stimulant options like atomoxetine or guanfacine.
2. **Frequent interrupts, lack of attentiveness, and recurrent behavioral problems** → Suggests ADHD or severe hyperkinetic disorder → Evaluate for need for behavioral modification or medication.
3. **Family history of Graves' disease (father)** → Consider thyroid function as a potential cause of hyperactivity or irritability (e.g., subclinical hyperthyroidism or mucopolysaccharidosis).
4. **History of otitis media** → Explore for secondary hearing impairment, as this could contribute to behavioral issues due to misinterpreted stimuli or discomfort from hearing loss.

Reasoning Process:

1. **Symptoms Analysis**:
 - The child exhibits hyperactivity, poor concentration, frequent interrupts, and lack of attentiveness. These symptoms are highly characteristic of **attention-deficit/hyperactivity disorder (ADHD)** or a related hyperkinetic disorder.
 - The recurrent episodes of behavioral issues, including hitting others and disrupting classes, further support a diagnosis of ADHD, as this condition often includes impulsive and inattentive symptoms.
2. **Thyroid Function**:
 - The father's history of Graves' disease raises the possibility of subclinical hyperthyroidism or another thyroid-related condition. However, there is no mention of symptoms such as weight loss, heat intolerance, palpitations, or tachycardia, which are common in hyperthyroidism. Additionally, the child's age (5 years) and lack of other symptoms (e.g., irritability, or sleep disturbances) make a primary thyroid disorder less likely.
 - Subclinical hyperthyroidism could contribute to hyperactivity and irritability, but without further evidence (e.g., elevated thyroid-stimulating hormone (TSH) or thyroid hormone levels), this remains a less likely explanation.
3. **Otitis Media**:
 - The child has a history of otitis media, but this is unlikely to be the primary cause of his current symptoms. However, hearing impairment could contribute to misinterpreting social cues or environmental stimuli, potentially exacerbating behavioral issues. Further evaluation for hearing impairment or related conditions is warranted.
4. **Developmental and Neurological Considerations**:
 - The child has met all developmental milestones, ruling out developmental delay or a significant neurological disorder as the cause of his symptoms. ADHD is the most appropriate diagnosis based on the behavioral history and exclusion of other conditions.
5. **Treatment Options**:
 - ADHD is commonly treated with **stimulant medications** (e.g., methylphenidate, amphetamine) or **non-stimulant medications** (e.g., atomoxetine, guanfacine).
 - Stimulants are first-line treatments for ADHD due to their efficacy in reducing hyperactivity, inattention, and impulsivity.
 - Non-stimulant options (e.g., atomoxetine, guanfacine) are considered if stimulants are contraindicated or if the child has comorbid conditions that contraindicate stimulant use (e.g., sleep apnea, anxiety).

Conclusion:

The most appropriate next step in treatment is **behavior therapy** combined with a **stimulant medication** (e.g., methylphenidate).

